AI READINESS 4.0

BUILD 5

Design a 10-Step Prompt Chain

Python Basics Study Guide and Quiz

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Course	IMCA — 2nd Year
Real task	Create beginner-friendly Python Basics notes, examples and quiz
Tools	ChatGPT, Perplexity, browser/source pages, Python interpreter, student review

Truthful run-status note: This document provides a prepared chain and illustrative trimmed outputs. The ten prompts have not all been executed in separate tools during preparation. Replace illustrative outputs with actual results as you run each step, and never mark a step complete unless you really performed it.

Includes ten stages, prompts, tools, trimmed outputs, two check steps, a human checkpoint, five-line reflection, run log, and reusable prompts.

Task Overview

Goal: Create a reliable, beginner-friendly Python Basics study guide for an IMCA second-year beginner.

Topics: running a program, variables, data types, input/output, operators, if/else, loops, and simple functions.

Workflow: define goal → plan topics → research → verify sources → draft notes → add examples → check code → create quiz → human review → finalise.

Ten-Step Prompt Chain

Step 1. Define the learning goal

Tool: ChatGPT

Prompt:
Role: You are a beginner-friendly Python instructor. Task: Define a realistic goal for a Python basics study guide. Context: I am an IMCA second-year beginner. Constraints: Keep it focused and achievable in one study session; assume no prior programming knowledge. Format: learning objective, topic list, and success criteria.

Trimmed output (illustrative until run): Illustrative: Cover variables, data types, input/output, operators, conditions, loops and simple functions. Success means I can explain each topic and solve a small question.

Step 2. Sequence the topics

Tool: ChatGPT

Prompt:
Role: You are a curriculum planner. Task: Put the selected Python topics in a logical order. Context: Use Step 1. Constraints: Start with prerequisites and explain the sequence. Format: numbered list with a short reason for each topic.

Trimmed output (illustrative until run): Illustrative: Start with running Python, then variables/types, input/output, operators, conditions, loops and functions.

Step 3. Research reliable sources

Tool: Perplexity

Prompt:
Role: You are a research assistant. Task: Find reliable beginner resources for Python basics. Context: I need sources for a student guide. Constraints: Prefer official Python documentation and reputable tutorials; provide direct links and never invent citations. Format: title, URL, topics covered, usefulness.

Trimmed output (illustrative until run): Illustrative: A shortlist of official/reputable resources with direct URLs. Record the actual URLs returned when you run this prompt.

Step 4. Verify sources — CHECK 1

Tool: Browser: open each source; Perplexity for discovery

Prompt:
Role: You are a source verifier. Task: Open and check each source from Step 3. Context: Only checked sources should support the guide. Constraints: Mark verified only if the page opens and supports the stated topic; remove broken or irrelevant sources. Format: source, opened yes/no, topic confirmed, keep/remove.

Trimmed output (illustrative until run): Illustrative: Keep only pages actually opened and confirmed relevant. Search snippets alone are not verification.

Check: Open every URL and record whether it loads and supports the topic.

Step 5. Draft study notes

Tool: ChatGPT with verified source list

Prompt:
Role: You are an experienced Python teacher. Task: Write notes on Python basics using the verified sources and topic sequence. Context: IMCA second-year beginner. Constraints: Define technical terms, use correct small examples, explain output, flag uncertainty. Format: headings, explanations, code examples and recap.

Trimmed output (illustrative until run): Illustrative: Notes on variables, int/float/str/bool, print(), input(), operators, if/else, loops and functions.

Step 6. Create worked examples

Tool: ChatGPT

Prompt:
Role: You are a Python lab instructor. Task: Make one short worked example for each core topic. Context: Use Step 5. Constraints:

Python 3, beginner syntax, runnable code, explain output, avoid advanced features. Format: topic, code, expected output, explanation.

Trimmed output (illustrative until run): Illustrative: Greeting input/output, if/else pass check, number loop and a function that adds two numbers.

Step 7. Check technical accuracy — CHECK 2

Tool: ChatGPT plus a Python interpreter

Prompt:
Role: You are a careful code reviewer. Task: Check the draft's code, explanations and expected outputs. Context: Review this draft, not a new guide. Constraints: Identify errors explicitly; do not claim code was executed unless it was. Format: issue, correction, test, corrected code.

Trimmed output (illustrative until run): Illustrative: An error list and corrections. Actually execute examples in Python where possible; AI review alone does not prove code runs.

Check: Execute code where possible and compare actual output with the explanation.

Step 8. Create quiz and answer key

Tool: ChatGPT

Prompt:
Role: You are a Python exam question setter. Task: Create 10 beginner questions from the final notes. Context: IMCA beginner revision. Constraints: Cover all core topics; each MCQ has one best answer; explain the key. Format: 10 questions followed by answers and brief explanations.

Trimmed output (illustrative until run): Illustrative: Ten questions on types, input/output, comparisons, conditions, loops and function calls, followed by an answer key.

Step 9. Human checkpoint — YOU review

Tool: Student (Adarsh); optional ChatGPT feedback

Prompt:
Role: Act as a review checklist, not a replacement for my judgement. Task: Help me review the study guide. Context: I will read the notes, run examples and attempt the quiz. Constraints: Do not mark this step complete for me. Format: checklist for clarity, correctness, usefulness, difficulty and gaps.

Trimmed output (illustrative until run): Complete personally: notes read [yes/no]; code examples run [list]; quiz score [score]; corrections [details]; weak topics [topics].

Human checkpoint: You must personally read, test, answer and record your review.

Step 10. Finalise the study kit

Tool: ChatGPT for editing; student for approval

Prompt:
Role: You are an editor preparing a study handout. Task: Organise checked notes, examples, quiz, answer key and sources. Context: Use reviewed content from Steps 1–9. Constraints: Do not add unverified facts; preserve source links and label unchecked items. Format: title, objectives, notes, examples, quiz, answer key, sources and reflection.

Trimmed output (illustrative until run): Illustrative: A structured Python Basics study kit with sources and any remaining checks clearly identified.

Run Log — Fill In While Executing

The outputs above are examples, not proof of execution. After running each prompt, record the actual tool, date and a short summary or correction.

Step	Tool actually used	Date	Actual output / issue
1	Not run yet	[date]	[fill after running]
2	Not run yet	[date]	[fill after running]
3	Not run yet	[date]	[fill after running]
4	Not run yet	[date]	[fill after running]
5	Not run yet	[date]	[fill after running]
6	Not run yet	[date]	[fill after running]
7	Not run yet	[date]	[fill after running]
8	Not run yet	[date]	[fill after running]
9	Not run yet	[date]	[fill after running]
10	Not run yet	[date]	[fill after running]

Suggested sources to verify

Resource	Direct URL	Verification
Official Python Tutorial	https://docs.python.org/3/tutorial/	Open and record yes/no
Python built-in functions	https://docs.python.org/3/library/functions.html	Open and record yes/no

Five-Line Reflection

1. The chain is designed to produce structured Python Basics notes, worked examples, a quiz, an answer key and a source list.

2. Breaking the task into ten stages makes planning and review easier than accepting one long AI answer.

3. Source verification and code checking matter because AI-generated content can contain errors.

4. The human checkpoint makes sure I personally understand the notes, test the code and identify weak topics.

5. I would improve the chain by recording actual run results, checking multiple reliable sources and revising the guide based on quiz mistakes.

Best Prompts for My Personal Prompt Playbook

Reusable Prompt A — Explain a concept

Role: You are an experienced programming teacher. Task: Explain [PYTHON TOPIC]. Context: I am an IMCA second-year beginner. Constraints: Use simple English, define technical words, include one runnable example and output, and avoid advanced details. Format: definition, explanation, code, output, common mistake and three practice questions.

Reusable Prompt B — Review code

Role: You are a careful Python code reviewer. Task: Check [PASTE CODE] and its expected output. Context: I am learning Python basics. Constraints: Identify errors explicitly, explain why, do not assume code was executed, and suggest a test. Format: issues, corrected code, expected output and verification steps.

Reusable Prompt C — Quiz with feedback

Role: You are a beginner Python tutor. Task: Test me on [TOPIC]. Context: I have just studied this topic. Constraints: Ask one question at a time and do not reveal the answer before I respond. Format: question, my response, feedback and next question.

Final Submission Checklist

- ■ Run all ten prompts and update the log truthfully.
- ■ Open source links and record whether they work and are relevant.
- ■ Run code examples in a Python interpreter when possible.
- ■ Complete the human checkpoint yourself.
- ■ Replace illustrative outputs with trimmed real outputs.
- ■ Add the reusable prompts to your Personal Prompt Playbook.
- ■ Submit through the AI Readiness 4.0 WhatsApp group form and select Build 5: 10-step prompt chain.